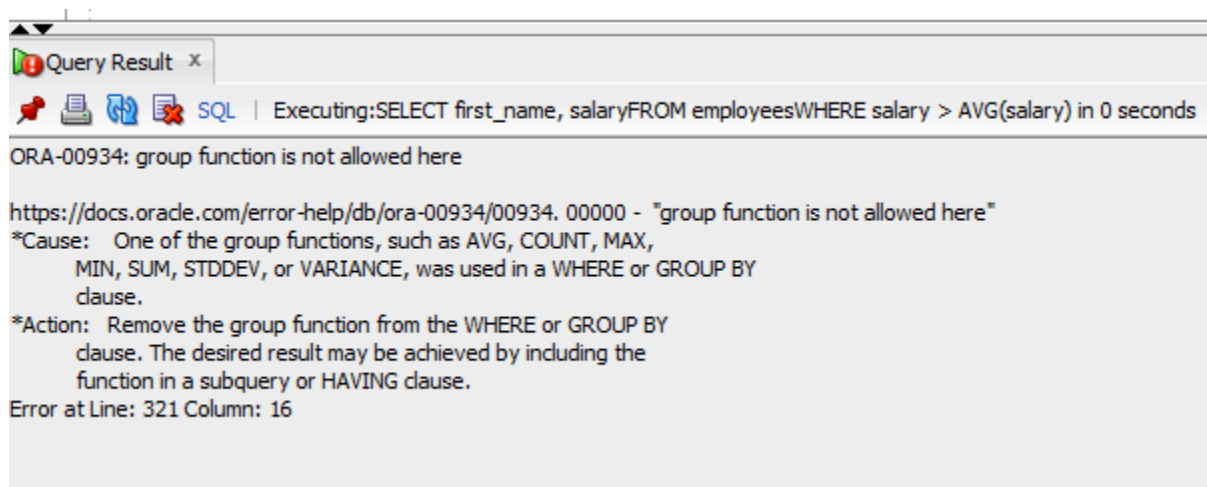


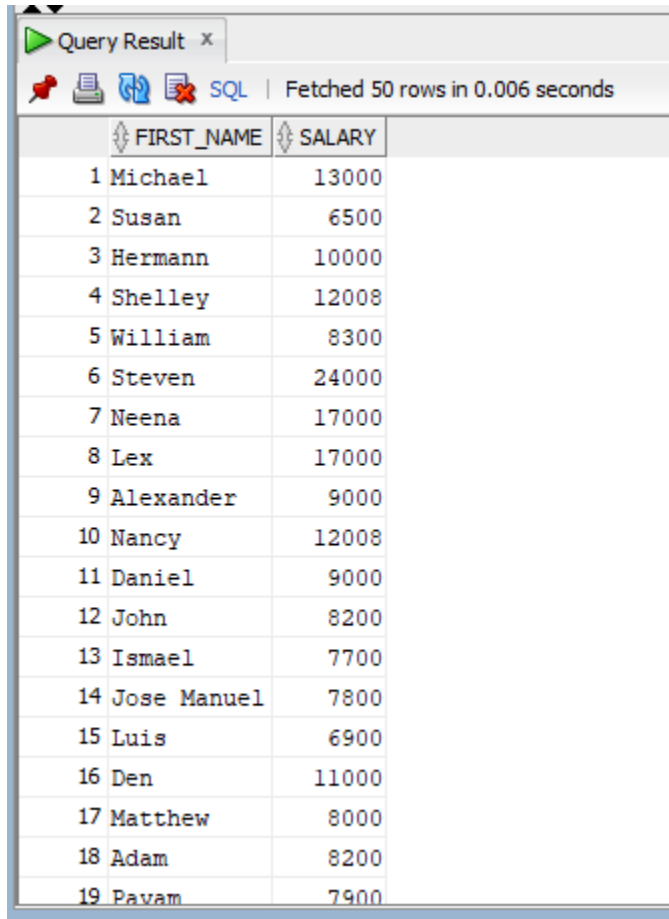
- **Query Review 1**

SELECT first_name, salary FROM employees WHERE salary > AVG(salary);



- **Analysis & Error:** The database cannot evaluate AVG(salary) in the WHERE clause because the WHERE clause filters rows *before* any aggregation takes place.
- **Business Logic Check:** the BL is incorrect, it will compare employee salary against the average before the average has been calculated.
- **Correction:** use subquery to compute the average.

SELECT first_name, salary FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);

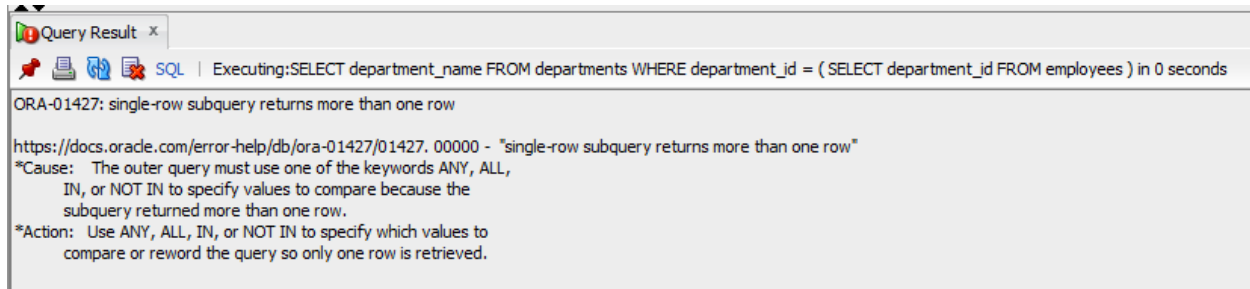


The screenshot shows a 'Query Result' window with a toolbar containing icons for pinning, printing, refreshing, and deleting, along with an 'SQL' label and a status message 'Fetched 50 rows in 0.006 seconds'. The table below displays the results of the query, listing 19 employees with their first names and salaries. The salaries are sorted in descending order, with Michael having the highest salary at 13,000 and Pavam having the lowest at 7,900. The window title is 'Query Result x'.

	FIRST_NAME	SALARY
1	Michael	13000
2	Susan	6500
3	Hermann	10000
4	Shelley	12008
5	William	8300
6	Steven	24000
7	Neena	17000
8	Lex	17000
9	Alexander	9000
10	Nancy	12008
11	Daniel	9000
12	John	8200
13	Ismael	7700
14	Jose Manuel	7800
15	Luis	6900
16	Den	11000
17	Matthew	8000
18	Adam	8200
19	Pavam	7900

- **Query Review 2**

SELECT department_name FROM departments WHERE department_id = (SELECT department_id FROM employees);



- **Analysis & Error:** This query used scalar subquery which expect one single row result, but when running the query it expect more than one rows which is incorrect.
- **Business Logic Check:** False. It looks for departments with employees but employs the incorrect operator.
- **Correction:** Change the single-row operator = to the **Multi-Row Operator IN**.

SELECT department_name FROM departments WHERE department_id IN (SELECT department_id FROM employees WHERE department_id IS NOT NULL);

The screenshot shows an Oracle SQL Developer window titled "Query Result x". The status bar indicates "All Rows Fetched: 11 in 0.005 seconds". Below the status bar, a table is displayed with the following data:

	DEPARTMENT_NAME
1	Administration
2	Marketing
3	Purchasing
4	Human Resources
5	Shipping
6	IT
7	Public Relations
8	Sales
9	Executive
10	Finance
11	Accounting

- **Query Review 3**

SELECT * FROM employees WHERE department_id IN (SELECT salary FROM employees);

The screenshot shows a SQL query result window titled 'Query Result'. The status bar indicates 'All Rows Fetched: 0 in 0.008 seconds'. The column headers are: EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, SALARY, COMMISSION_PCT, MANAGER_ID, and DEPARTMENT_ID. The result grid is empty.

- **Analysis & Error:** when running the query, it don't show an output.
- **Business Logic Check:** It checks whether an employee's department_id matches anyone's salary . It will return empty.
- **Correction:** Align the filtering column of the outer query with the SELECT column of the subquery.

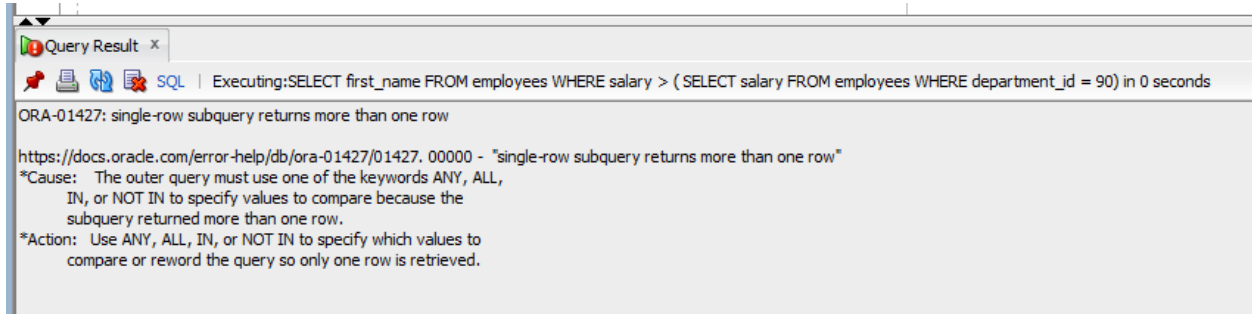
SELECT * FROM employees WHERE department_id IN (SELECT DISTINCT department_id FROM employees WHERE department_id IS NOT NULL);

The screenshot shows a SQL query result window titled 'Query Result'. The status bar indicates 'Fetched 50 rows in 0.008 seconds'. The column headers are: EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, SALARY, COMMISSION_PCT, MANAGER_ID, and DEPARTMENT_ID. The result grid contains 19 rows of employee data.

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	198 Donald	OConnell	DOCONNEL	650.507.9833	21-JUN-07	SH_CLERK	2600	(null)	124	50
2	199 Douglas	Grant	DGRANT	650.507.9844	13-JAN-08	SH_CLERK	2600	(null)	124	50
3	200 Jennifer	Whalen	JWHALEN	515.123.4444	17-SEP-03	AD_ASST	4400	(null)	101	10
4	201 Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100	20
5	202 Pat	Fay	PFAY	603.123.6666	17-AUG-05	MK_REP	6000	(null)	201	20
6	203 Susan	Mavris	SMAVRIS	515.123.7777	07-JUN-02	HR_REP	6500	(null)	101	40
7	204 Hermann	Baer	HBAER	515.123.8888	07-JUN-02	PR_REP	10000	(null)	101	70
8	205 Shelley	Higgins	SHIGGINS	515.123.8080	07-JUN-02	AC_MGR	12008	(null)	101	110
9	206 William	Gietz	WGIEZT	515.123.8181	07-JUN-02	AC_ACCOUNT	8300	(null)	205	110
10	100 Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
11	101 Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
12	102 Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
13	103 Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
14	104 Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
15	105 David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(null)	103	60
16	106 Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4800	(null)	103	60
17	107 Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60
18	108 Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
19	109 Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100

- **Query Review 4**

```
SELECT first_name FROM employees WHERE salary > ( SELECT salary FROM employees
WHERE department_id = 90);
```



- **Analysis & Error:** It will run fine *only if* Department 90 has exactly one employee. If Department 90 has multiple employees it will crash with error.
- **Business Logic Check:** Do we want employees to earn more than the highest or lowest earner in Department 90?
- **Correction:** Use a multi-row quantification operator.